Zscaler Cloud Connector - Function App Settings Restore

Manual procedure for when the lifecycle Function App has lost its application settings. Written 10 November 2026.

Applies to	Names shown	Shell	Status
zscct, modules/function-app	Prod, westus2 (ccprdarm...uw2001)	PowerShell with Azure CLI	Not yet run against a live subscription

A. Before you start

If you can run Terraform, stop here and run `terraform apply` from the environment folder instead. The module writes every setting in this document, so an apply puts them back in one pass. Carry on only if an apply is not possible.

#	Confirm
A1	Azure CLI is signed in to the subscription that holds the Function App.
A2	You can write app settings on the Function App. Reader alone shows the settings as empty even when they are present.
A3	To hand: resource group name, storage account name, Key Vault URI, connector provisioning URL, and `short_id` from tfvars.
A4	The names below are prod. For dev or slz, change the `prd` token to match.
A5	You are looking at the right app. Check the resource group and that there is no deployment slot selected.

B. Collect the values

1 Set the two values only you know, and the three fixed names. The storage account starts `ccprdarmstuw2001` followed by the first four characters of `short_id`.

```
$RG   = "<resource-group>"
$ST   = "<storage-account-name>"
$APP  = "ccprdarmfuncuw2001"
$MI   = "ccprdarmmifuncuw2001"
$APPI = "ccprdarmappiuw2001"
```

2 Look up the rest.

```
$SUB   = az account show --query id -o tsv
$MICID = az identity show -g $RG -n $MI --query clientId -o tsv
$MIID  = az identity show -g $RG -n $MI --query id -o tsv
$AI    = az monitor app-insights component show -g $RG --app $APPI `
           --query connectionString -o tsv
$BLOB  = az storage blob list --account-name $ST -c function-packages `
           --auth-mode login --query "[0].name" -o tsv
```

3 Print them. None should be blank.

```
$SUB; $MICID; $MIID; $AI; $BLOB
```

`$BLOB` is the one most likely to come back empty, because the storage firewall only admits the Function App subnet and the deployer address. If it does, build the name yourself. It is `zscc-func-<short_id>-<md5 of the zip>.zip`. The hash of the zip on Zscaler's main branch today is `afea19813d47369cba6a0b95a51bc15d`; for any other copy, compute it.

```
$MD5  = (Get-FileHash "<path>\zscaler_cc_function_app.zip" -Algorithm MD5).Hash.ToLower()
$BLOB = "zscc-func-<short_id>-$MD5.zip"
```

C. Apply the settings

A backtick continues the line in PowerShell. It has to be the last character on the line, with nothing after it, not even a space.

4 Fill in the Key Vault URI and the provisioning URL, then run. The provisioning URL goes in without `https://`.

```
az functionapp config appsettings set -g $RG -n $APP --settings `
  "SUBSCRIPTION_ID=/subscriptions/$SUB" `
  "MANAGED_IDENTITY=$MICID" `
  "RESOURCE_GROUP=$RG" `
  "VAULT_URL=<key-vault-uri>" `
  "CC_URL=<provisioning-url-without-https://>" `
  "TERMINATE_UNHEALTHY_INSTANCES=true" `
  "WEBSITE_RUN_FROM_PACKAGE=https://$ST.blob.core.windows.net/function-packages/$BLOB" `
  "WEBSITE_RUN_FROM_PACKAGE_BLOB_MI_RESOURCE_ID=$MIID" `
  "AzureWebJobsStorage__accountName=$ST" `
  "AzureWebJobsStorage__credential=managedidentity" `
  "AzureWebJobsStorage__clientId=$MICID" `
  "APPLICATIONINSIGHTS_CONNECTION_STRING=$AI" `
  "ApplicationInsightsAgent_EXTENSION_VERSION=~3" `
  "XDT_MicrosoftApplicationInsights_Mode=recommended" `
  "FUNCTIONS_EXTENSION_VERSION=~4" `
  "FUNCTIONS_WORKER_RUNTIME=python"
```

5 `VMSS_NAME` is a JSON list and PowerShell drops the inner double quotes, so it gets its own command. One name per zone. On a Windows machine:

```
az functionapp config appsettings set -g $RG -n $APP `
  --settings 'VMSS_NAME=[\"ccprdarmvms-z1-uw2001\",\"ccprdarmvms-z2-uw2001\"]'
```

In Azure Cloud Shell the backslashes are not wanted:

```
az functionapp config appsettings set -g $RG -n $APP `
  --settings 'VMSS_NAME=["ccprdarmvms-z1-uw2001","ccprdarmvms-z2-uw2001"]'
```

6 Read it back. It must print the list with its double quotes and no backslashes. If not, run step 5 again using the other form.

```
az functionapp config appsettings list -g $RG -n $APP `
  --query "[?name=='VMSS_NAME'].value" -o tsv
```

Expected: `["ccprdarmvms-z1-uw2001","ccprdarmvms-z2-uw2001"]`

D. Restart and sync

7 Restart the app.

```
az functionapp restart -g $RG -n $APP
```

8 Sync the triggers. Until this is done the app does not load its package and the timers do not fire. Terraform makes the same call.

```
$URL = "https://management.azure.com/subscriptions/$SUB/resourceGroups/$RG" +
       "/providers/Microsoft.Web/sites/$APP/syncfunctiontriggers?api-version=2024-04-01"
az rest --method post --url $URL
```

E. Verify

9 List the functions. There should be two.

```
az functionapp function list -g $RG -n $APP --query "[].name" -o tsv
```

Check	What you are looking for
Two functions listed	The health monitor, on a one minute timer, and the resource sync, every thirty minutes.
Invocations	In the portal, the health monitor shows a successful run within a few minutes of the sync.
Terraform agrees	Next time someone can run `terraform plan`, it reports no change to the Function App settings. Any difference it shows is a value typed wrongly here.

F. Reference

What each setting should hold, taken from `modules/function-app/locals.tf` and the provider defaults. The names are fixed by the vendor package.

Setting	Value	Note
SUBSCRIPTION_ID	/subscriptions/...	Full path, not the bare GUID
MANAGED_IDENTITY	client id	Function identity, ccprdarmmifuncuw2001
RESOURCE_GROUP	resource group name	
VMSS_NAME	["<z1>","<z2>"]	JSON list, one scale set per zone
VAULT_URL	vault URI	From the Key Vault overview page
CC_URL	provisioning URL	No https://. The package adds it
TERMINATE_UNHEALTHY_INSTANCES	true	false only if disable_health_remediation is true
WEBSITE_RUN_FROM_PACKAGE	blob URL	Container function-packages
WEBSITE_RUN_FROM_PACKAGE_BLOB_MI_RESOURCE_ID	identity resource id	Full /subscriptions/... path
AzureWebJobsStorage__accountName	storage account name	Set by the provider
AzureWebJobsStorage__credential	managedidentity	
AzureWebJobsStorage__clientId	client id	Same as MANAGED_IDENTITY
APPLICATIONINSIGHTS_CONNECTION_STRING	connection string	Set by the provider from site_config
ApplicationInsightsAgent_EXTENSION_VERSION	~3	
XDT_MicrosoftApplicationInsights_Mode	recommended	
FUNCTIONS_EXTENSION_VERSION	~4	Set by the provider
FUNCTIONS_WORKER_RUNTIME	python	Set by the provider

G. Notes

No storage connection string. There is no `AzureWebJobsStorage` setting on this app and one must not be added. Shared keys are switched off on the storage account, and the host reaches it as the managed identity through the three `AzureWebJobsStorage__` settings.

Health check path. Leave it blank. The module does not set one. Both functions run on timers and nothing answers on HTTP, so a probe would have nothing to call.

How the settings might have been lost. The repository does not say which of these happened. An apply that failed part way, after the app was created. Another deployment or script that wrote app settings and replaced the whole list. An Azure Policy that rejected or changed the update. Or they were never lost, and the account looking has Reader only.